Approval-Based Elections in the 1D Euclidean Domain

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TODO

0.1 10/12/2020

1. Refactor code - vcr ilp detection, profile generation
2. Generate and analyze profiles: 5x5, 4x6, 6x4, random NxM
3. Implement ilp MAV
4. Implement VR/CR MAV Liu and Guo (2016)
5. Compare ilp and poly MAV

0.2 17/12/2020

1.

Abstract

Praca polega na opracowaniu algorytmów dla szeregu problemów wyborczych w jednowymiarowej dziedzinie euklidesowej (w modelu Voter/Candidate Range). Oprócz algorytmów, dla problemów NP-trudnych w większych dziedzinach preferencji, w ramach projektu będzie należało także:

- poszukiwać algorytmów rozpoznających jednowymiarowe dziedziny Euklidesowe na podstawie danych aprobatowych
- oszacować jak wiele wyborów należy do rozważanej dziedziny
- poszukiwać algorytmów stwierdzających jak podobne są dane wybory do pochodzących z dziedziny jednowymiarowej

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Chapter 1

Introduction

1.1 Section One

1.2 Section Two

Chapter 2

Preliminaries

2.1 Papers

- An Analysis of Approval-Based Committee Rules for 2D-Euclidean Elections by Faliszewski (2020)
- Overview of various domain restriction in Structure in Dichotomous Preferences by Elkind and Lackner (2015)
- Minimax Approval Voting rule introduced by Brams et al. (2007)
- Polynomial algorithms for MAV in VR and CR domains by Liu and Guo (2016)
- Example of a new domain restriction (SPOC) and recognition algorithms by Peters and Lackner (2020)
- Example of a NP-hard domain recognition algorithms and theorems by Peters (2017)
- Alternative approch to C1P for domain recognition by Magiera and Faliszewski (2019)

2.2 Definitions

Approval Elections Election is a pair $E = (C,V)$ where $C = \{c_1, ..., c_m\}$ is a set of candidates and $V = \{v_1, ..., v_n\}$ is a list of voters. Each voter $v \in V$ approves a subset of the candidates and disapproves the remaining candidates.

Preference Restrictions

One-Dimensional Euclidean Preferences Elkind and Lackner (2015)

Voter Candidate Range Given an election $E = (C, V)$, we say that the voters have *1D-Euclidean* preferences For each agent there exists a point x_a and nonnegative realvalue r_a such that: $c \in A(v) \Leftrightarrow \sqrt{(x_c - x_v)^2} \leq r_c + r_v$

Intuitevely, for $a \in C \cup V$ the point x_a describes a's ideal position in 1-dimmensional space of opinions. For candidate $c \in C, r_c$ can be seen as c's charisma. For candidate $v \in V, r_v$ can be seen as v's willingness to compromise.

Candidate Range We say that P satisfies CR if candidates in C can be ordered so that each of the sets vi forms an interval of that ordering.

CR is special case of VCR, where we require all the voters to have radii equal to zero

Voter Range We say that P satisfies VR if the voters in P can be reordered so that for every candidate c the voters that approve c form an interval of that ordering.

VR is special case of VCR, where we require all the candidates to have radii equal to zero

The VCR model is strictly more powerful than the VR and CR ones. For example, election with candidate set $C = \{a, b, c, d\}$ and voters with approval sets $\{a, b\}$, $\{b, c\}$, $\{b, d\}$, and $\{a, b, c, d\}$ is VCR, but neither VR nor CR.

Committee Selection Rules

Approval Voting

Proportional Approval Voting

Minimax Approval Voting (MAV) Brams et al. (2007)

As close as possible to the preferences of all parties. By "close" we mean that the maximum distance of the compromise from the position of any state is minimal, which we call a minimax outcome.

Selects a committee W that minimizes the maximum Hamming distance between W and voters' approval ballots. We define MAV-score of a committee W as $Max(f(W, v1), f(W, v2), ..., f(W, vn))$ where $f(W, vi) =$. MAV returns a committee W with the lowest MAV-score.

Chapter 3

Approval-Based Elections in the *1D* Euclidean Domain

3.1 Section One

Chapter 4

Solution

4.1 VCR Profiles

4.1.1 How many VCR profiles exists that are not VR nor CR

3C 3V

- all profiles $2^9 = 512$
- VCR ILP detection 506
- VCR not VR nor CR 0

4C 4V

- all profiles $2^{16} = 65536$
- VCR ILP detection 57832
- VCR not VR nor CR 96

5C 5V

- all profiles $2^{25} = 33554432$

6C 4V
4C 6V
Medium Size
Large Size

4.2 Poly VCR Profile Detection Algorithm

4.2.1 Integer Linear Programming

The only input is *ApprovalSet* of each voter.
Represented by $2D$ matrix A_{vc}

Variables

x_c - candidate c position on x axis
x_v - voter v position on x axis
r_c - candidate c radius
r_v - voter v radius

Transform VCR condition to Linear constraints

$$\begin{aligned} VCR = & \\ &= \sqrt{(x_c - x_v)^2} \leq r_c + r_v \\ &= \|x_c - x_v\| \leq r_c + r_v \\ &= x_c - x_v \leq r_c + r_v \; or - (x_c - x_v) \leq r_c + r_v \\ &= -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \end{aligned} \tag{4.1}$$

Positive and Negative Condition and Big M method

Introduce large constant positive parmater
M

Introduce new binary variables for each pair of candidate and voter:
$z1_{cv}$
$z2_{cv}$

$$
\begin{aligned}
&\textit{Minimize 0}\\
&\textit{foreach (c,v) in } C \times V:\\
&\textit{if } c \in ApprovalSet(v) \textit{ add constraints}\\
&x_c - x_v \leq r_c + r_v\\
&-(x_c - x_v) \leq r_c + r_v\\
&\textit{if } c \notin ApprovalSet(v) \textit{ add constraints}\\
&x_c - x_v + M(1 - z1_{cv}) \geq r_c + r_v + 1\\
&-(x_c - x_v) + M(1 - z2_{cv}) \geq r_c + r_v + 1\\
&z1_{cv} + z2_{cv} >= 1
\end{aligned}
$$

Mock Distance Variable Condition

Introduce new variable
D_{cv} - distance between candidate c and voter v on x axis

$$
\begin{aligned}
&\textit{Minimize 0}\\
&\textit{foreach (c,v) in } C \times V:\\
&\textit{add constraints}\\
&D_{cv} = x_c - x_v\\
&D_{cv} = -(x_c - x_v)\\
&\textit{if } c \in ApprovalSet(v) \textit{ add constraints}\\
&D_{cv} \leq r_c + r_v\\
&\textit{if } c \notin ApprovalSet(v) \textit{ add constraints}\\
&D_{cv} \geq r_c + r_v + 1
\end{aligned}
$$

4.3 Poly Minimax Algorithm for VCR profiles

4.4 Approval deletion for VCR profile to become VR or CR

Chapter 5

Evaluation

5.1 Section One

5.2 Section Two

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